

An Analysis of Ensemble learning

Ensemble learning

Ensemble learning -- Part 1

Overview Supervised learning algorithms search through a hypothesis space to find a suitable hypothesis that will make good predictions with a particular problem. Even if this space contains hypotheses that are very well-suited for a particular problem, it may be very difficult to find a good one. Ensembles combine multiple hypotheses to form one which should be theoretically better. Ensemble learning trains two or more machine learning algorithms on a specific classification or regression task. The algorithms within the ensemble model are generally referred as "base models", "base learners", or "weak learners" in literature. These base models can be constructed using a single modelling algorithm, or several different algorithms. The idea is to train a diverse set of weak models on the same modelling task, such that the outputs of each weak learner have poor predictive ability (i.e., high bias), and among all weak learners, the outcome and error values exhibit high variance. Fundamentally, an ensemble learning model trains at least two high-bias (weak) and high-variance (diverse) models to be combined into a better-performing model. The set of weak models — which would not produce satisfactory predictive results individually — are combined or averaged to produce a single, high performing, accurate, and low-variance model to fit the task as required. Ensemble learning typically refers to bagging (bootstrap aggregating), boosting or stacking/blending techniques to induce high variance among the base models. Bagging creates diversity by generating random samples from the training observations and fitting the same model to each different sample — also known as homogeneous parallel ensembles. Boosting follows an iterative process by sequentially training each base model on the up-weighted errors of the previous base model, producing an additive model to reduce the final model errors — also known as sequential ensemble learning. Stacking or blending consists of different base models, each trained independently (i.e. diverse/high variance) to be combined into the ensemble model — producing a heterogeneous parallel ensemble. Common applications of ensemble learning include random forests (an extension of bagging), Boosted Tree models, and Gradient Boosted Tree Models. Models in applications of stacking are generally more task-specific — such as combining clustering techniques with other parametric and/or non-parametric techniques. Evaluating the prediction of an ensemble typically requires more computation than evaluating the prediction of a single model. In one sense, ensemble learning may be thought of as a way to compensate for poor learning algorithms by performing a lot of extra computation. On the other hand, the alternative is to do a lot more learning with one non-ensemble model. An ensemble may be more efficient at improving overall accuracy for the same increase in compute,

storage, or communication resources by using that increase on two or more methods, than would have been improved by increasing resource use for a single method. Fast algorithms such as decision trees are commonly used in ensemble methods (e.g., random forests), although slower algorithms can benefit from ensemble techniques as well. By analogy, ensemble techniques have been used also in unsupervised learning scenarios, for example in consensus clustering or in anomaly detection.

Ensemble learning -- Part 2

where y is the predicted class, C is the set of all possible classes, H is the hypothesis space, P refers to a probability, and T is the training data. As an ensemble, the Bayes optimal classifier represents a hypothesis that is not necessarily in H . The hypothesis represented by the Bayes optimal classifier, however, is the optimal hypothesis in ensemble space (the space of all possible ensembles consisting only of hypotheses in H). This formula can be restated using Bayes' theorem, which says that the posterior is proportional to the likelihood times the prior:

Ensemble learning -- Part 3

In statistics and machine learning, ensemble methods use multiple learning algorithms to obtain better predictive performance than could be obtained from any of the constituent learning algorithms alone. Unlike a statistical ensemble in statistical mechanics, which is usually infinite, a machine learning ensemble consists of only a concrete finite set of alternative models, but typically allows for much more flexible structure to exist among those alternatives.

Ensemble learning -- Part 4

Change detection Change detection is an image analysis problem, consisting of the identification of places where the land cover has changed over time. Change detection is widely used in fields such as urban growth, forest and vegetation dynamics, land use and disaster monitoring. The earliest applications of ensemble classifiers in change detection are designed with the majority voting, Bayesian model averaging, and the maximum posterior probability. Given the growth of satellite data over time, the past decade sees more use of time series methods for continuous change detection from image stacks. One example is a Bayesian ensemble changepoint detection method called BEAST, with the software available as a package Rbeast in R, Python, and Matlab.

Ensemble learning -- Part 5

An Analysis of Ensemble learning

Ensemble learning

Face recognition, which recently has become one of the most popular research areas of pattern recognition, copes with identification or verification of a person by their digital images. Hierarchical ensembles based on Gabor Fisher classifier and independent component analysis preprocessing techniques are some of the earliest ensembles employed in this field.

Ensemble learning -- Part 6

Bootstrap aggregating (bagging) Bootstrap aggregation (bagging) involves training an ensemble on bootstrapped data sets. A bootstrapped set is created by selecting from original training data set with replacement. Thus, a bootstrap set may contain a given example zero, one, or multiple times. Ensemble members can also have limits on the features (e.g., nodes of a decision tree), to encourage exploring of diverse features. The variance of local information in the bootstrap sets and feature considerations promote diversity in the ensemble, and can strengthen the ensemble. To reduce overfitting, a member can be validated using the out-of-bag set (the examples that are not in its bootstrap set). Inference is done by voting of predictions of ensemble members, called aggregation. It is illustrated below with an ensemble of four decision trees. The query example is classified by each tree. Because three of the four predict the positive class, the ensemble's overall classification is positive. Random forests like the one shown are a common application of bagging.